

Summary

Electrical engineering PhD researcher with Cornell and Yale ECE training across thin-film fabrication, semiconductor physics, integrated photonics, device simulation, and scientific computing. Experience spans cleanroom GaN device development, optical and materials characterization, light-matter modeling, and distributed HPC software for semiconductor and photonic systems.

Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Fabricated GaN power-electronics test structures: deposited GaN by MOCVD; defined contacts by e-beam evaporation and photolithography; patterned trenches; and used ion-implantation-based selective-area etching.
- Characterized thin films and fabricated structures using SEM, SEM cathodoluminescence, and Raman spectroscopy; connected process choices to optical and structural observations.
- Simulated high- k -dielectric MOSFETs in Silvaco; modeled VCSEL light-matter physics with coupled photon, phonon, and electron dynamics.
- Built distributed workflows for exciton-polarons and optical materials using Quantum ESPRESSO, post-DFT many-body perturbation methods, CUDA/ROCm, MPI, OpenMP, PETSc, and SLEPc.
- Used agentic ML tooling to generate first-principles software inputs from journal literature and technical documentation.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and delivered APS presentations in 2024, 2025, and 2026 on self-trapped exciton formation and related excited-state materials physics.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Parallelized CUDA/C++ software for analyzing and visualizing acoustic and nuclear well-tool data used in engineering diagnostics and field interpretation.
- Applied DOLFINx/PETSc finite-element analysis to frequency studies of steel pipes in well environments.
- Supported instrumentation-oriented engineering around deployed well tools, connecting field data, sensor systems, and numerical analysis.

FindLight

Photonics Technical Writer

2018–2019

- Wrote technical content on lasers, integrated optics, and optoelectronic components for engineering-facing audiences.

Selected Projects

1550 nm SiN Band-Pass Filter: Designed a Meep FDTD band-pass filter for the 1550 nm telecom band; fabricated SiN waveguides and measured the filter response with an erbium-doped fiber laser.

High- k MOSFET Simulation: Used Silvaco device simulation to analyze MOSFET electrostatics and transport with high- k gate dielectrics.

Cadence VLSI / Analog Design: Designed a multi-layer-perceptron circuit in Cadence and designed an analog voltage-controlled oscillator (VCO), applying transistor-level design and verification workflows.

DOLFINx Finite-Element Analysis: Built 2D DOLFINx/PETSc simulations for transient heat diffusion, Poisson potential around a circular cutout, and incompressible flow past an immersed cylinder, with PyVista and Matplotlib results.

VCSEL Light-Matter Modeling: Built simulation tools for VCSEL emission and coupled photon, phonon, and electron interactions in semiconductor systems.

Equivariant Scientific ML: Developed PyTorch Geometric equivariant neural networks for material energy prediction and optical-property prediction.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University*MS, Electrical Engineering*

2023

Cornell University*MEng, Electrical and Computer Engineering*

2017–2018

Cornell University*BS, Electrical and Computer Engineering*

2013–2017

Technical Skills

Fabrication / Characterization: MOCVD GaN deposition, e-beam contacts, photolithography, trench patterning, ion implantation, selective-area etching, SEM, SEM-CL, Raman spectroscopy, laser-bench testing

Devices / Photonics: Semiconductor physics, thin films, integrated photonics, lasers, VCSEL modeling, light-matter interaction

Simulation / Design: Silvaco, Cadence, Meep FDTD, DOLFINx, Quantum ESPRESSO, semiconductor device simulation, finite-element analysis, and post-DFT many-body perturbation theory

Machine Learning: PyTorch, PyTorch Geometric, equivariant graph neural networks, optical- and materials-property prediction

Software / HPC: Python, C, C++, CUDA, ROCm/HIP, MPI, OpenMP, PETSc, SLEPc, DOLFINx, Linux, Docker

GPU / FDTD: cuBLAS, cuTENSOR, cuRAND, rocBLAS, rocFFT, Meep FDTD, cuda-gdb, nvprof